



■ PLANRS ACADEMY

PERIMETER OF IRREGULAR SHAPES

A Complete Lesson Guide with Examples, Activities & Assessment

■ Mathematics Geometry &
Measurement

■ Session Duration 30
Minutes

■ Session 1 of 1 Beginner
Level

■ NCERT / CBSE Aligned

"Perimeter = the total distance around the outside. Add ALL the sides — even if they are all different!"

LESSON OVERVIEW
Session 1 — Perimeter of Irregular Shapes

WHAT THIS LESSON IS ABOUT

COURSE OVERVIEW

This lesson introduces learners to the concept of **perimeter** as applied to **irregular shapes** — shapes that do not have all sides equal in length. Rather than memorising formulas, learners will physically trace the boundary of familiar real-world objects, measure each side, and add all the side lengths together to find the total distance around the shape.

The session is rooted in real life: the boundary of a *khet* (field), the edge of a *rangoli* design, the fence around a garden — all irregular shapes that learners encounter every day. By the end of this session, learners will understand that **perimeter is simply the total distance around the outside of a shape**, regardless of whether all sides are the same or different.

■ WHY THIS TOPIC MATTERS

Builds additive reasoning (adding multiple quantities)
Develops spatial awareness and boundary thinking
Creates physical intuition: 'walk around the outside'
Confidence with measurement units (cm and m)
Foundation for area, fencing, and real-world geometry

■ PRIOR KNOWLEDGE ASSUMED

Can identify basic 2D shapes (square, rectangle, triangle)
Can measure a straight line with a ruler in centimetres
Can add 2–4 single or double-digit numbers
Understands the concept of 'side' on a shape
No prior knowledge of perimeter or formula required

LEARNING OBJECTIVES

LO	BLOOM'S LEVEL	OBJECTIVE
LO 1	Remember	Define perimeter as the total distance around the outside boundary of any shape.
LO 2	Understand	Explain that irregular shapes have sides of different lengths; perimeter = sum of ALL sides, not a formula.
LO 3	Apply	Measure each side of a given irregular shape with a ruler and calculate the perimeter by adding all side lengths.
LO 4	Apply / Analyse	Identify and describe at least two real-life Indian examples where perimeter is relevant (khet, rangoli, cricket ground).
LO 5	Create	Trace an irregular shape of own choice, measure each side, and calculate the perimeter independently.

KEY CONCEPTS

The building blocks of this lesson

CONCEPT 1

What Is Perimeter?

Perimeter is the **total distance** you would walk if you went all the way **around the outside** of a shape and came back to where you started.



It is the **total length of all the sides added together**.

Core Idea: Perimeter = Walk Around the Outside = Add ALL Sides

■ THE ANT WALK ANALOGY

Imagine a tiny ant starting at one corner of a shape. The ant walks along **every side** — never going inside, never lifting its feet — and comes back to where it started. The total distance the ant walked is the **perimeter**!



REAL-LIFE INDIAN EXAMPLES

■ Khet (Farm Field)

A farmer walking around the boundary of their field to check crops — the total distance walked is the perimeter.

■ Rangoli Design

If you trace your finger along the outer boundary of a Diwali rangoli, the total length of that trace is the perimeter.

■ Cricket Ground

Walking around the outside of a cricket ground for exercise — you are walking the perimeter of the ground!

CONCEPT 2

What Is an Irregular Shape?

REGULAR Shape

All sides are the **same length**.

Example: Square, equilateral triangle

You **CAN** use a formula: $P = n \times \text{side}$

VS

IRREGULAR Shape

Sides are **different lengths**.

Example: L-shaped room, khet field

You **MUST** add every side individually!

THE 4-STEP METHOD

A repeatable process that works for ANY irregular shape

No matter how complicated an irregular shape looks, the method to find its perimeter is always the same. Follow these four steps every time:

1**LABEL EVERY SIDE**

Mark each side with a letter (A, B, C...) — even the short ones. Missing a side = wrong answer!

2**WRITE THE LENGTH OF EACH SIDE**

Record the measurement of each labelled side. Use your ruler, or read from the diagram.

3**ADD ALL SIDE LENGTHS TOGETHER**

Add every single measurement. The ant walks ALL sides — so you add ALL numbers.

4**WRITE THE ANSWER WITH A UNIT**

Always include the unit (cm or m). A number without a unit is an incomplete answer.

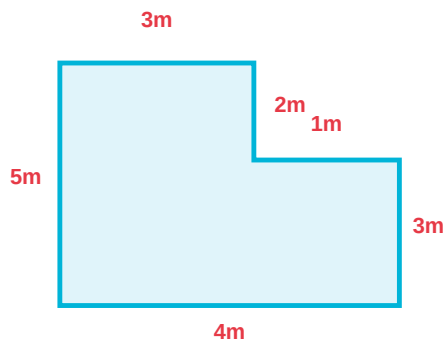
■ GOLDEN RULE

For **IRREGULAR** shapes: there is **NO** shortcut formula. You **MUST** add every single side individually. A shape can have 4 sides, 6 sides, or 10 sides — the method never changes. **LABEL → ADD → UNIT.**

Hindi Connect: **Paridhi** (■■■■■) — for bilingual classrooms | Pronunciation: **puh-RIM-i-ter** | Unit: Centimetre = **Sentimeetar** (■■■■■■■■■■) | Metre = **Meetar** (■■■■)

WORKED EXAMPLES

Step-by-step solutions with full working shown

WORKED EXAMPLE 1**L-Shaped Veranda**

Shape: An L-shaped veranda with 6 sides.

STEP 1 — Label sides:

Side A = 5m | Side B = 3m | Side C = 2m

Side D = 1m | Side E = 3m | Side F = 4m

STEP 2 — Write each length:

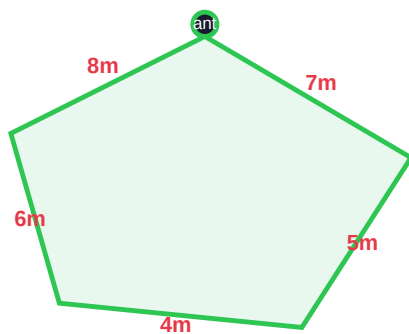
5m, 3m, 2m, 1m, 3m, 4m

STEP 3 — Add all sides:

$$5 + 3 + 2 + 1 + 3 + 4 = 18$$

STEP 4 — Write with unit:

Perimeter = 18 metres

WORKED EXAMPLE 2**Kishan's Khet Field**

Shape: Kishan's 5-sided khet (field).

*Problem: Kishan needs to fence his khet. How many metres of fencing does he need?***STEP 1 — Label sides:**

Side A = 8m | Side B = 6m | Side C = 4m

Side D = 5m | Side E = 7m

STEP 3 — Add all sides:

$$8 + 6 + 4 + 5 + 7 = 30$$

Kishan needs 30 metres of fencing!

GUIDED PRACTICE

Activities to consolidate understanding through doing

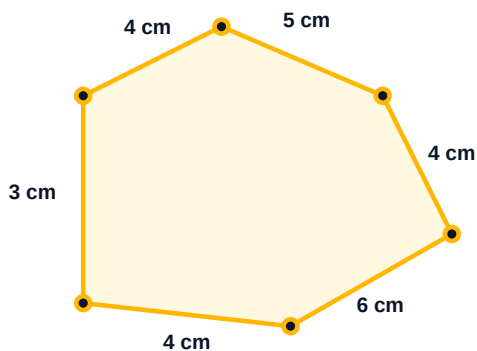
THE STRING METHOD

Hands-On Measurement Technique**How to use the String (Dhaga) Method:**

1. Place a cotton thread (dhaga) along the boundary of any irregular shape, following every side exactly.
2. Cut the string at the point where it meets the starting corner.
3. Straighten the string and measure its total length with a ruler.
4. **That measurement is the perimeter!** This is especially useful for shapes that are hard to label individually.

Materials needed: Cotton thread (dhaga), ruler, scissors.

PRACTICE ACTIVITY

Find the Perimeter of This Shape**Using the 4-Step Method:**

STEP 1 — All 6 sides are labelled on the shape. Start at any corner and work your way around systematically.

STEP 2 — Side lengths are given in centimetres (cm):

A = 3 cm | B = 4 cm | C = 5 cm**D = 4 cm | E = 6 cm | F = 4 cm****STEP 3 — Add all sides:**

$$3 + 4 + 5 + 4 + 6 + 4 = ?$$

Answer: Perimeter = 26 cm

DISCUSSION QUESTIONS

- Can you name other irregular shapes you see at home or in your neighbourhood?
- Why do we need to know the perimeter of a field or garden? Who uses this in real life?
- How does measuring with string help when you can't use a ruler directly on a curved shape?
- What happens if you forget to measure one side? Will your perimeter be correct?
- Can two shapes have the same perimeter but look completely different? How?

COMMON MISCONCEPTIONS & TEACHING TIPS

Know what learners get wrong — and how to fix it

MISCONCEPTION 1

■ WHAT LEARNERS THINK:

Perimeter is the space inside the shape.

■ REALITY:

Perimeter is the distance AROUND the OUTSIDE — the boundary, not the interior. The space inside is called area (introduced later).

■ TEACHING TIP:

Use the 'ant walk' — the ant stays on the OUTSIDE. It never goes inside. Total ant walk = perimeter. Physical activity: trace the OUTSIDE of your hand.

MISCONCEPTION 2

■ WHAT LEARNERS THINK:

For an irregular shape, I multiply one side by the number of sides.

■ REALITY:

This only works for REGULAR shapes where all sides are equal. For irregular shapes, you MUST add each side individually.

■ TEACHING TIP:

Show a shape where multiplying gives the wrong answer. 'If I multiplied this one side by 4, I'd get the wrong answer — because the other sides are different!'

MISCONCEPTION 3

■ WHAT LEARNERS THINK:

I only need to count the sides I can see clearly — the obvious ones.

■ REALITY:

Every side must be included — even short ones, even ones at an angle. Missing even one side gives a wrong perimeter.

■ TEACHING TIP:

The ant walk is the cure: 'The ant walks along EVERY side. It cannot skip any. If it skips one, it doesn't make it back to start!'

MISCONCEPTION 4

■ WHAT LEARNERS THINK:

Perimeter is always in centimetres.

■ REALITY:

Perimeter uses the same unit as the side measurements. If sides are in metres, perimeter is in metres.

■ TEACHING TIP:

'We always use the same unit as the sides. If the farmer measures his khet in metres, the perimeter is in metres — not centimetres.'

MISCONCEPTION 5

■ WHAT LEARNERS THINK:

A shape with more sides always has a bigger perimeter.

■ REALITY:

Perimeter depends on the LENGTH of sides, not just the number. A triangle with sides $10+10+10 = 30$ cm can be larger than a hexagon with sides $1+1+1+1+1+1 = 6$ cm.

■ TEACHING TIP:

Show two contrasting examples. 'Which has a bigger perimeter — the big triangle or the tiny hexagon? Let's calculate and find out!'

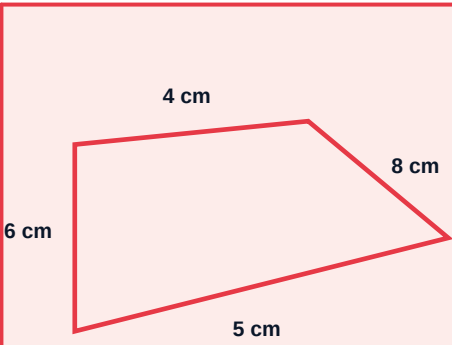
KEY VOCABULARY

Words every learner must know and be able to use

PERIMETER	(puh-RIM-i-ter)	Paridhi (■■■■■)
The total distance around the outside boundary of any shape. Found by adding all side lengths together.		
IRREGULAR SHAPE	(i-REG-yoo-ler SHAPE)	Visham Akriti (■■■■ ■■■■■)
A shape whose sides are NOT all equal in length. The perimeter must be found by adding each side individually.		
REGULAR SHAPE	(REG-yoo-ler SHAPE)	Sam Akriti (■■ ■■■■■)
A shape where ALL sides are equal in length (e.g., square, equilateral triangle). Can use a formula for perimeter.		
BOUNDARY	(BOUN-da-ree)	Seema (■■■■)
The outer edge or border of a shape. Perimeter measures the total length of the boundary.		
CENTIMETRE (cm)	(SEN-ti-mee-ter)	Sentimeetar (■■■■■■■■)
A unit of length. 100 centimetres = 1 metre. Used for shorter measurements like sides of small shapes.		
METRE (m)	(MEE-ter)	Meetar (■■■■)
A unit of length. 1 metre = 100 centimetres. Used for larger measurements like fields and rooms.		

EXIT TICKET

Quick formative assessment — to be completed individually



A quadrilateral with side lengths 4 cm, 8 cm, 5 cm, and 6 cm.

Find the perimeter of this shape.

All four sides are labelled. Use the 4-step method:

6 cm + 4 cm + 8 cm + 5 cm = ?

Write your answer here:

Perimeter = _____ cm

Answer: 6 + 4 + 8 + 5 = 23 cm

LESSON SUMMARY

The 3 big ideas from this session

1

Perimeter is the total distance around the OUTSIDE of a shape.

It is the sum of all side lengths. Not the inside area — the outer boundary.

2

Irregular shapes have sides of DIFFERENT lengths.

They are everywhere in real life — khet fields, rangoli, rooms, cricket grounds.

3

To find perimeter: Label every side → Add ALL lengths → Write the unit.

The method never changes, no matter how many sides the shape has.

SUCCESS INDICATORS

A learner has met all objectives when they can:

(a)	Look at any irregular shape with labelled sides and add them correctly to find the perimeter.
(b)	Say: "Perimeter is the total distance around the outside. I add all the sides — even if they are all different!"
(c)	Connect the idea to one real-life Indian example without prompting.

TAKE-HOME CHALLENGE



Find an irregular shape at home!

1. Trace around the outside of any irregular object (a book cover, a leaf, a room floor plan).
2. Measure ALL sides with a ruler or cotton string (dhaga).
3. Add all the lengths together carefully.
4. Write the total perimeter with the correct unit.
5. Bring your shape tracing and calculation to share next time!

WHAT COMES NEXT

→ Perimeter of regular shapes and introduction of formulas

→ Area of irregular shapes — comparing perimeter and area

→ Perimeter and area of combined shapes

→ Land measurement and real-world geometry problems

PLANRS ACADEMY | Mathematics — Geometry & Measurement

"Learning that feels like play — for every Indian child."